

# Zeki Duman

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## CONTACT INFORMATION

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## EDUCATION

**University of California Santa Barbara**, Santa Barbara, US **2018 - Current**  
**M.S./PhD**, *Electrical and Computer Engineering*

- Area: System Identification and Control, applied to pharmacokinetic modeling and closed-loop drug delivery
- GPA: 3.68/4.0
- Expected Graduation: December 2026

**Bogazici University**, Istanbul, Turkey **2013 - 2018**

**B.S.**, *Electrical and Electronics Engineering*, **Minor Economics**

- GPA: 3.91/4.0, Dean's High Honors List

**Columbia University in the City of New York**, New York, US **2017**

**B.S. Study Abroad**, *Electrical and Computer Engineering*

- GPA: 3.83/4.0

## RESEARCH EXPERIENCE

**Graduate Research Assistant**, UCSB, Santa Barbara, US **2022 - Present**

Institute for Collaborative Biotechnologies

Co-advisors: Prof. Joao P. Hespanha (ECE) and Prof. Kevin W. Plaxco (Chemistry)

### Pharmacokinetic Modeling and System Identification

- Designed compartmental pharmacokinetic models describing drug and metabolite concentration-time data in plasma and solid tissue from in-vivo animal studies.
- Developed system identification and parameter estimation methods using nonlinear programming in Python and MATLAB (CasADi, IPOPT) to fit PK models to high-frequency experimental data.

### Model-Informed Closed-Loop Drug Delivery

- Implemented model predictive control with moving horizon estimation of PK parameters to regulate methotrexate concentrations in real time in rats.
- Achieved rapid and accurate regulation of plasma and solid-tissue drug concentrations using adaptive and model predictive control (published in *ACS Pharmacology and Translational Science*, 2025).
- Developing a population-based adaptive control approach that maintains accurate drug regulation across subjects with diverse pharmacokinetics (manuscript in preparation).

### Signal Processing for Real-Time In-Vivo Drug Monitoring

- Developed peak extraction methods for square-wave voltammetry that reduce noise in drug concentration measurements from electrochemical aptamer-based (EAB) sensors.
- Designed Multi-Gaussian with Laplacian voltammogram fitting, improving the precision of seconds-resolved in-vivo drug measurements (published in *ACS Sensors*, 2025).
- Collaborated with chemists and life scientists to design experiments, define data requirements, and interpret results.

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| PUBLICATIONS      | <ul style="list-style-type: none"> <li>• <b>Z. Duman</b>, N.A. Emmons, T.E. Kippin, L. Sepunaru, J. Hespanha, K.W. Plaxco, “Improving Real-Time In Vivo Molecular Monitoring: Multi-Gaussian with Laplacian Voltammogram Fitting Enhances the Precision of Electrochemical Aptamer-Based Sensors”, <i>ACS Sensors</i>, May, 2025.</li> <li>• N.A. Emmons, <b>Z. Duman</b>, M.K. Erdal, J. Hespanha, T.E. Kippin, K.W. Plaxco, “Feedback control over plasma drug concentrations achieves rapid and accurate control over solid-tissue drug concentrations”, <i>ACS Pharmacology and Translational Science</i>, April, 2025.</li> <li>• J.M. Gibson, <b>Z. Duman</b>, N.A. Emmons, D.O. Sogbesan, J.P. Hespanha, C.B. Kirkpatrick, T.E. Kippin, K.W. Plaxco, “A Minimally Invasive, Subcutaneous Biosensor Reliably Estimates Plasma Methotrexate Exposure and Quantifies Plasma-to-Tissue Transport Kinetics”, <i>ACS Pharmacology and Translational Science</i>, May, 2026.</li> </ul> |                                                             |
| SKILLS            | <p><b>Programming Languages:</b> Python, MATLAB</p> <p><b>Pharmacokinetic Modeling:</b> Compartmental PK Modeling, Parameter Estimation, System Identification, ODE and State-Space Modeling, Inter-Subject Variability</p> <p><b>Control and Optimization:</b> Model Predictive Control, Moving Horizon Estimation, Adaptive Control, Nonlinear Programming</p> <p><b>Scientific Computing:</b> NumPy, SciPy, CasADi, IPOPT, Simulink</p> <p><b>Data Analysis:</b> Time-Series and Signal Processing, Curve Fitting, Noise Reduction</p> <p><b>Languages:</b> Turkish (Native), English (Fluent), Korean (Intermediate)</p>                                                                                                                                                                                                                                                                                                                                                            |                                                             |
| TEACHING          | <p><b>Teaching Assistant experience in UCSB:</b> Assisted lead professors with administering college-level courses, led problem-solving discussions and hardware labs, and explained technical material to students from varied backgrounds.</p> <ul style="list-style-type: none"> <li>– ECE 130A: Signals and Systems</li> <li>– ECE 147C: Advanced Control Laboratory</li> <li>– ECE 15A: Fundamentals of Logic Design</li> <li>– CS 8: Introduction to Computer Science (Python)</li> <li>– ECE 152A: Digital Design Principles</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |
| HONORS AND AWARDS | <p><b>Treasurer in ECE Graduate Student Association, UCSB</b></p> <p><b>ECE Outstanding Teaching Assistant Award, UCSB</b></p> <p><b>Ranked 24<sup>th</sup> in National University Entrance Exam, Turkey</b></p> <p><b>High School Valedictorian, Turkey</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>2023, 2024</p> <p>2021, 2022</p> <p>2013</p> <p>2013</p> |